HR Analytics:

Employee Attrition Intelligence

This project analyzes employee attrition using data-driven techniques to uncover key drivers of turnover and identify high-risk segments. It combines statistical analysis to support strategic HR decision-making.

What This Report Covers:

- Overall attrition trends and workforce distribution
- Impact of salary, overtime, and satisfaction on attrition
- Department-wise and role-based attrition patterns
- Tenure and promotion Gap and age group analysis
- Key insight to improving employee retention

Tools & Technologies:

- Python (Pandas, Matplotlib, Seaborn)
- Jupyter Notebook
- Data Analysis & Visualization

Objectives:

To Transform raw data into actionable business insights that helps business to reduce attrition and improve workforce stability.

12 Key Insights

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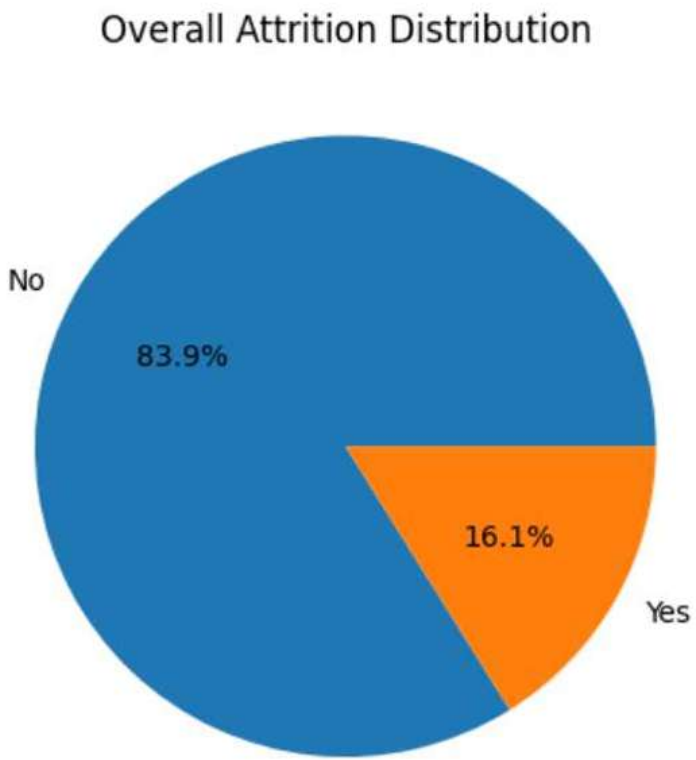
1. Overall Attrition Distribution.	**2.** Combined Satisfaction Score vs Attrition.	**3.** Promotion Gap vs Attrition.	**4.** Overall Attrition Distribution.
5. Overall Attrition Distribution.	**6.** Overall Attrition Distribution.	**7.** Overall Attrition Distribution.	**8.** Overall Attrition Distribution.
9. Overall Attrition Distribution.	**10.** Overall Attrition Distribution.	**11.** Overall Attrition Distribution.	**12.** Overall Attrition Distribution.

```python
attr = df['Attrition'].value_counts()

plt.figure()
attr.plot(kind='pie', autopct='%1.1f%%')
plt.title("Overall Attrition Distribution")
plt.ylabel('')
plt.show()
```

[15] Python

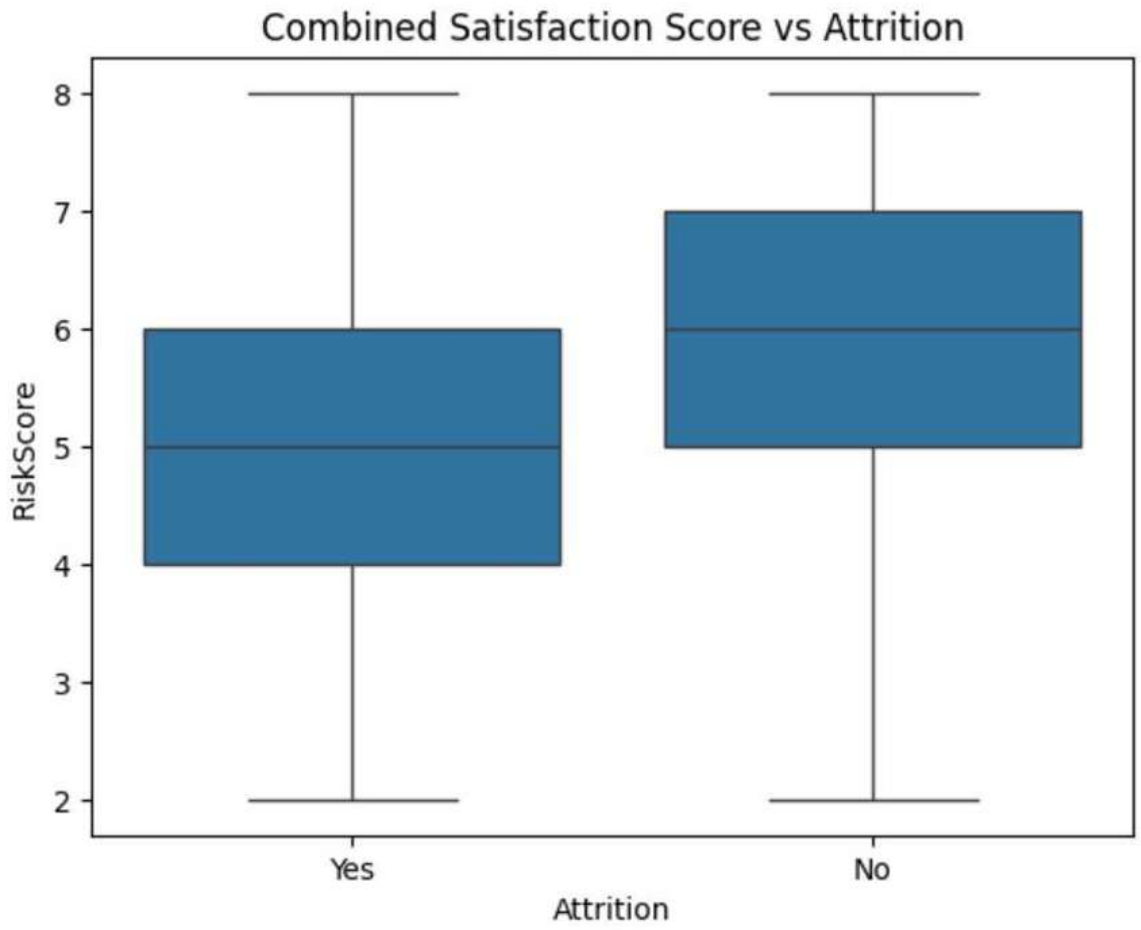


Insights:
1. Attrition rate at 16.1% reflects a moderate but meaningful level of employee turnover.
2. High retention (83.9%) indicates overall workforce stability with manageable risk.
3. The 16.1% rate serves as a benchmark to identify high-risk segments across the organization

```python
df['RiskScore'] = df['JobSatisfaction'] + df['WorkLifeBalance']

sns.boxplot(x='Attrition', y='RiskScore', data=df)
plt.title("Combined Satisfaction Score vs Attrition")
plt.show()
```

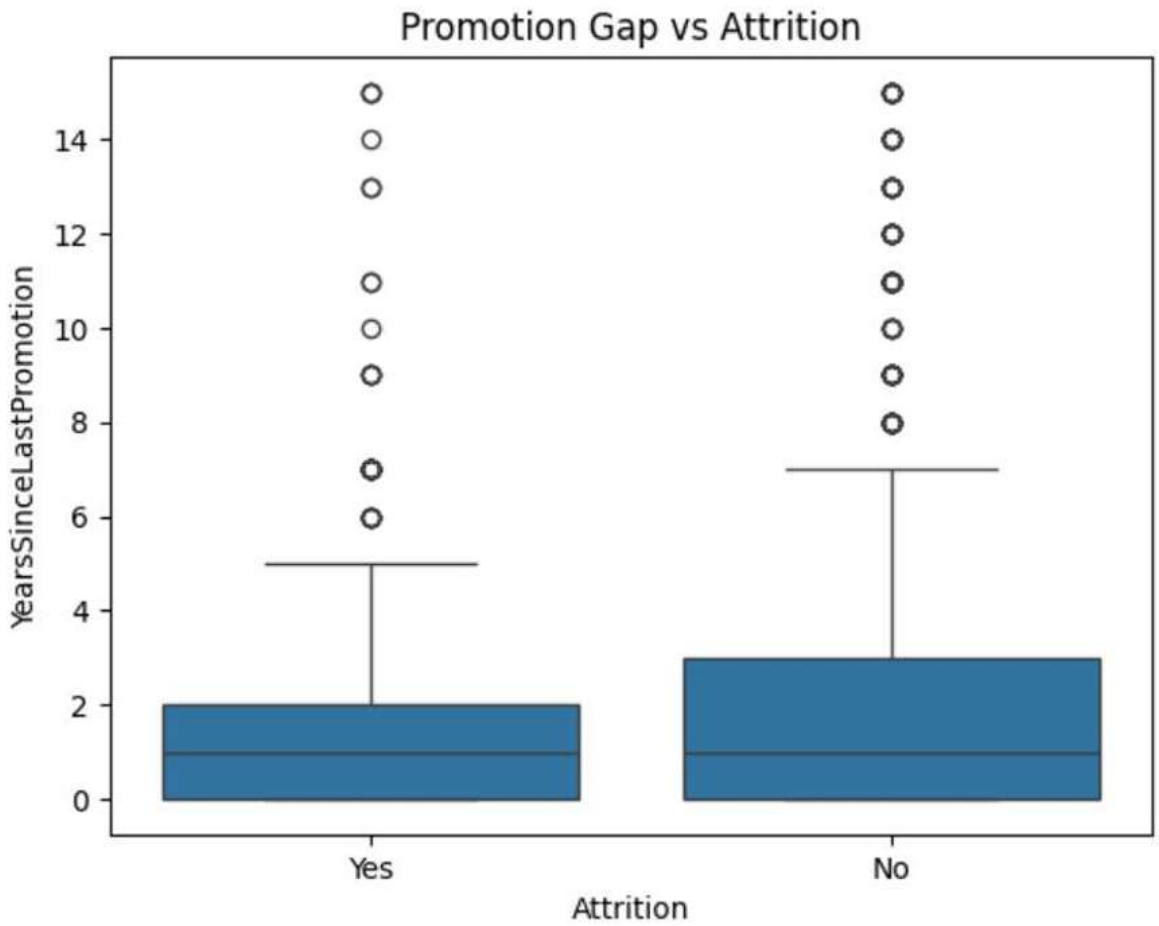
[26] Python



Insight:
A composite score of job satisfaction + work-life balance clearly separates leavers from stayers. Those who left had a median combined score of ~4.5 (out of 8), while stayers averaged ~6. This composite metric could serve as a simple early-warning indicator.

```python
sns.boxplot(x='Attrition', y='YearsSinceLastPromotion', data=df)
plt.title("Promotion Gap vs Attrition")
plt.show()
```

[25] Python



Insight:
Employees who left had a median of 3 years since last promotion, compared to 2 years for those who stayed. Promotion delays of 3+ years increase exit risk.

```python
plt.figure()
sns.histplot(data=df, x='YearsAtCompany', hue='Attrition', bins=15)
plt.title("Tenure vs Attrition")
plt.show()
```

[19] Python

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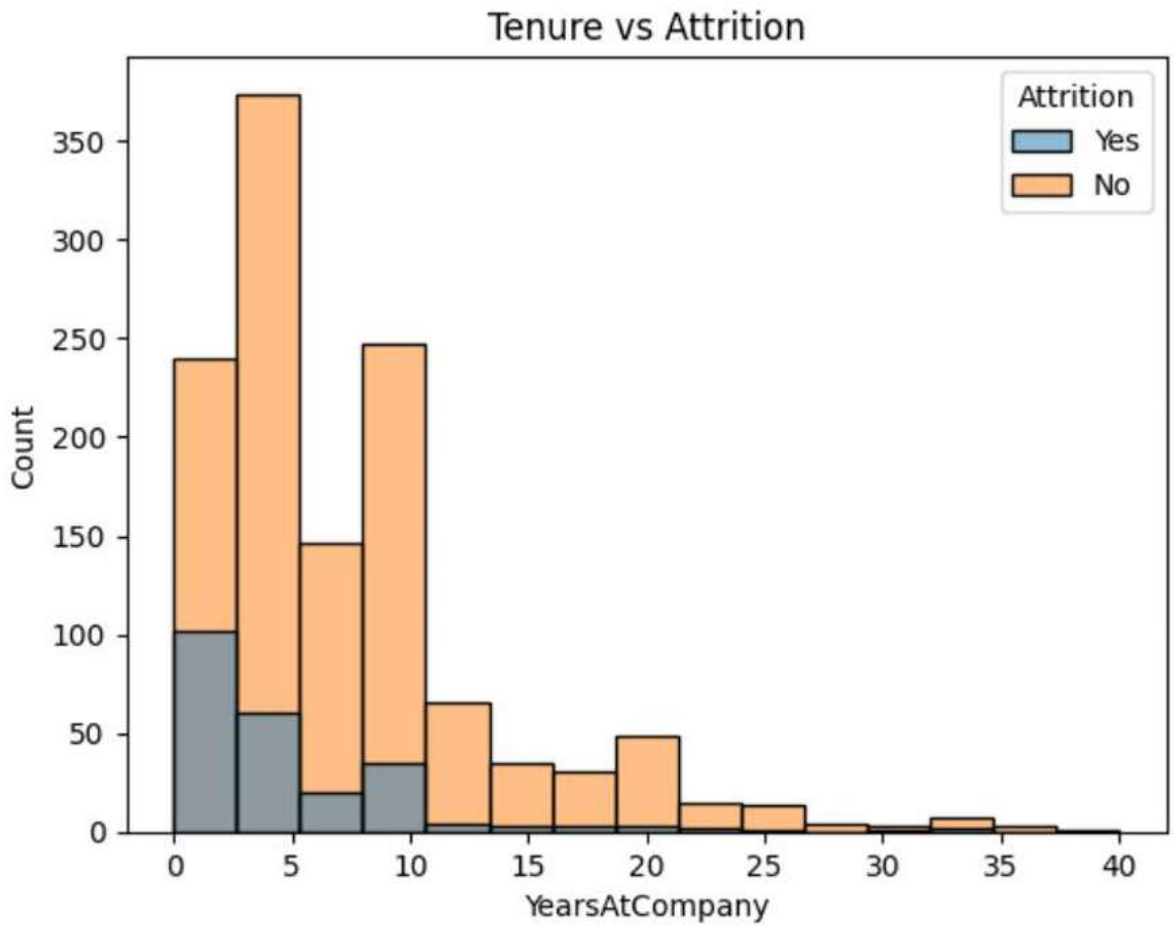
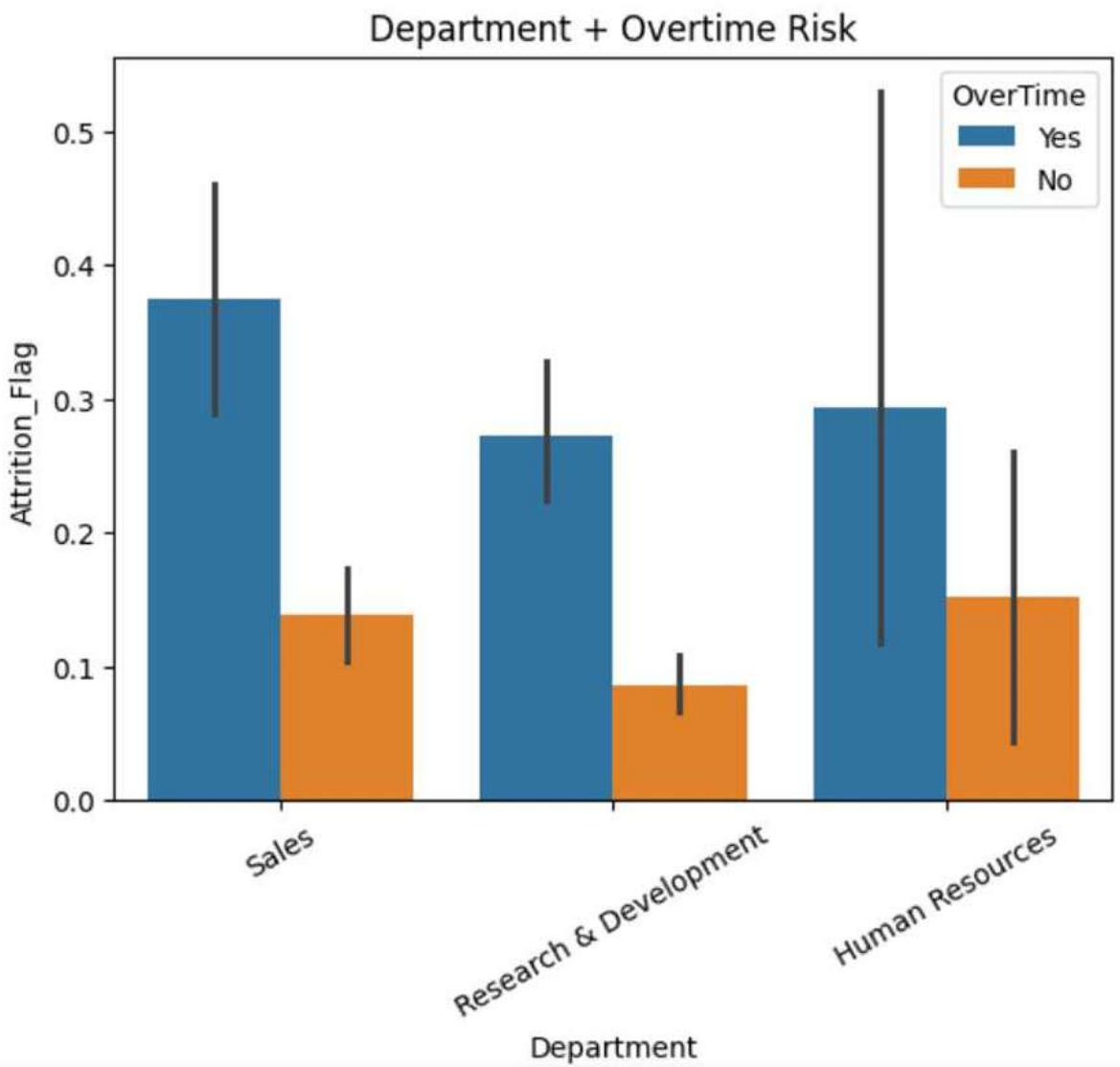


Insight:
Most exits occur within the first 5 years, with a peak in years 0–2. Long-tenure employees (7+ years) rarely leave. Early-career engagement and onboarding are critical.

```python
sns.barplot(x='Department', y='Attrition_Flag', hue='OverTime', data=df)
plt.xticks(rotation=30)
plt.title("Department + Overtime Risk")
plt.show()
```

[24] Python



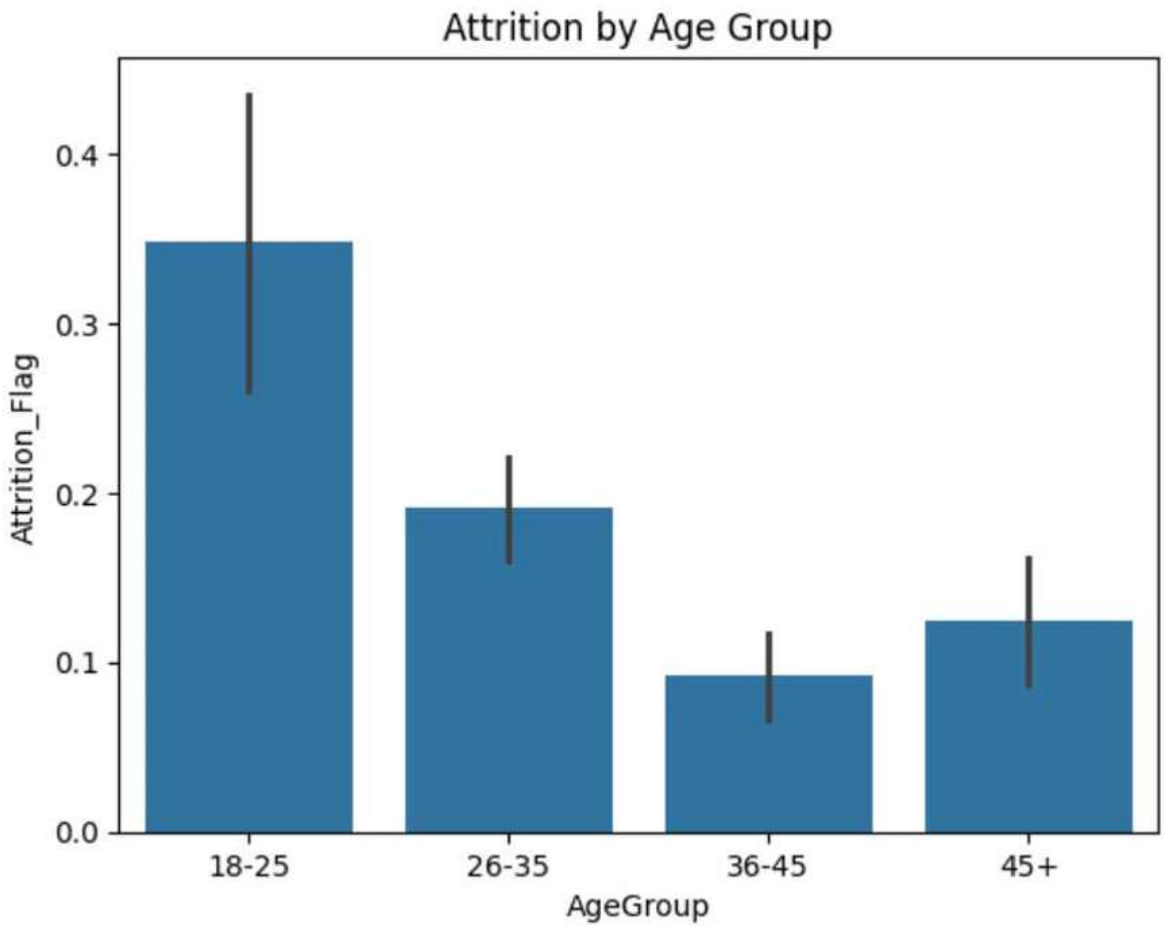
Insight

Overtime is the strongest driver of attrition across all departments, with Sales employees showing the highest risk due to workload pressure.

```python
df['AgeGroup'] = pd.cut(df['Age'], bins=[18,25,35,45,60],
                        labels=['18-25','26-35','36-45','45+'])

sns.barplot(x='AgeGroup', y='Attrition_Flag', data=df)
plt.title("Attrition by Age Group")
plt.show()
```

[23] Python



Insight:
Attrition is highest in the 26–35 age group (~30%) and 18–25 (~25%). Mid-career employees are most likely to leave — often due to growth stagnation or work-life imbalances.

```python
sns.barplot(x='WorkLifeBalance', y='Attrition_Flag', data=df)
plt.title("Work-Life Balance vs Attrition")
plt.show()
```

[21] Python

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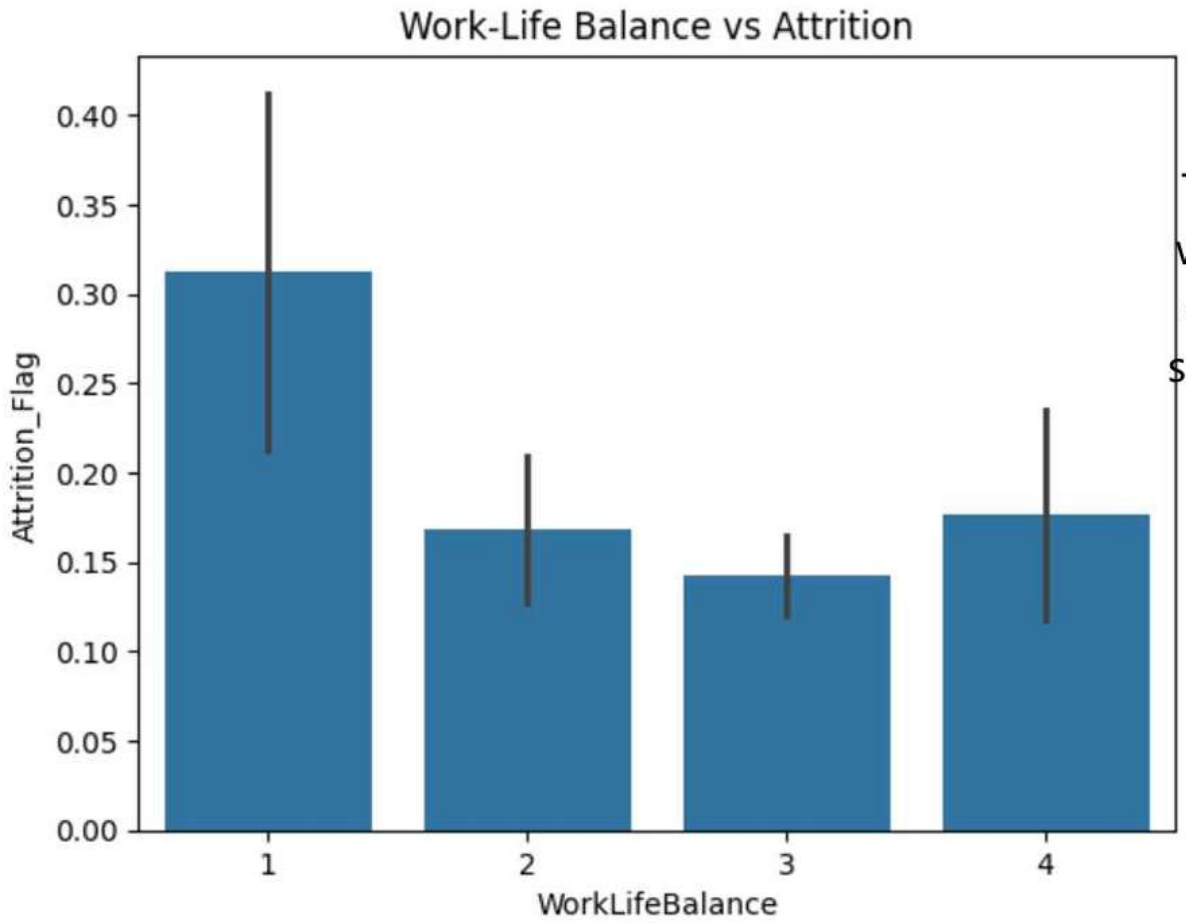

Insight:
The highest attrition occurs among employees who rated their work-life balance as “Good” (3) — not “Poor” or “Average.” This suggests that surface-level satisfaction can hide deeper issues like workload creep or stalled careers.

```python
role = df.groupby('JobRole')['Attrition_Flag'].mean().sort_values(ascending=False)*100

plt.figure(figsize=(8,5))
role.plot(kind='barh')
plt.title("Attrition by Job Role")
plt.show()
```

[20] Python

Insight:

Sales Representative tops the list at ~40% attrition, followed by Laboratory Technician (~24%) and HR (~23%). Role-specific retention plans are essential.

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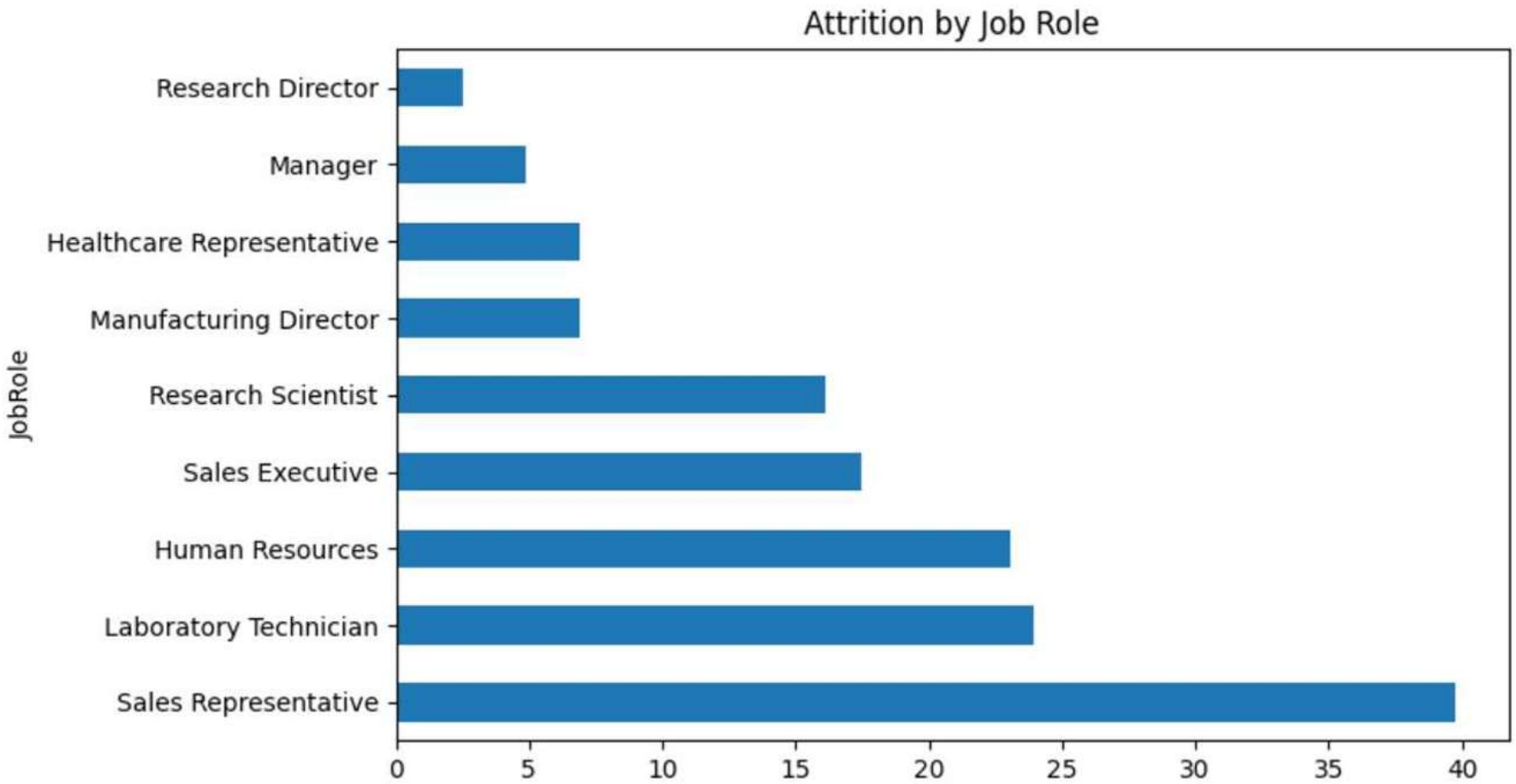
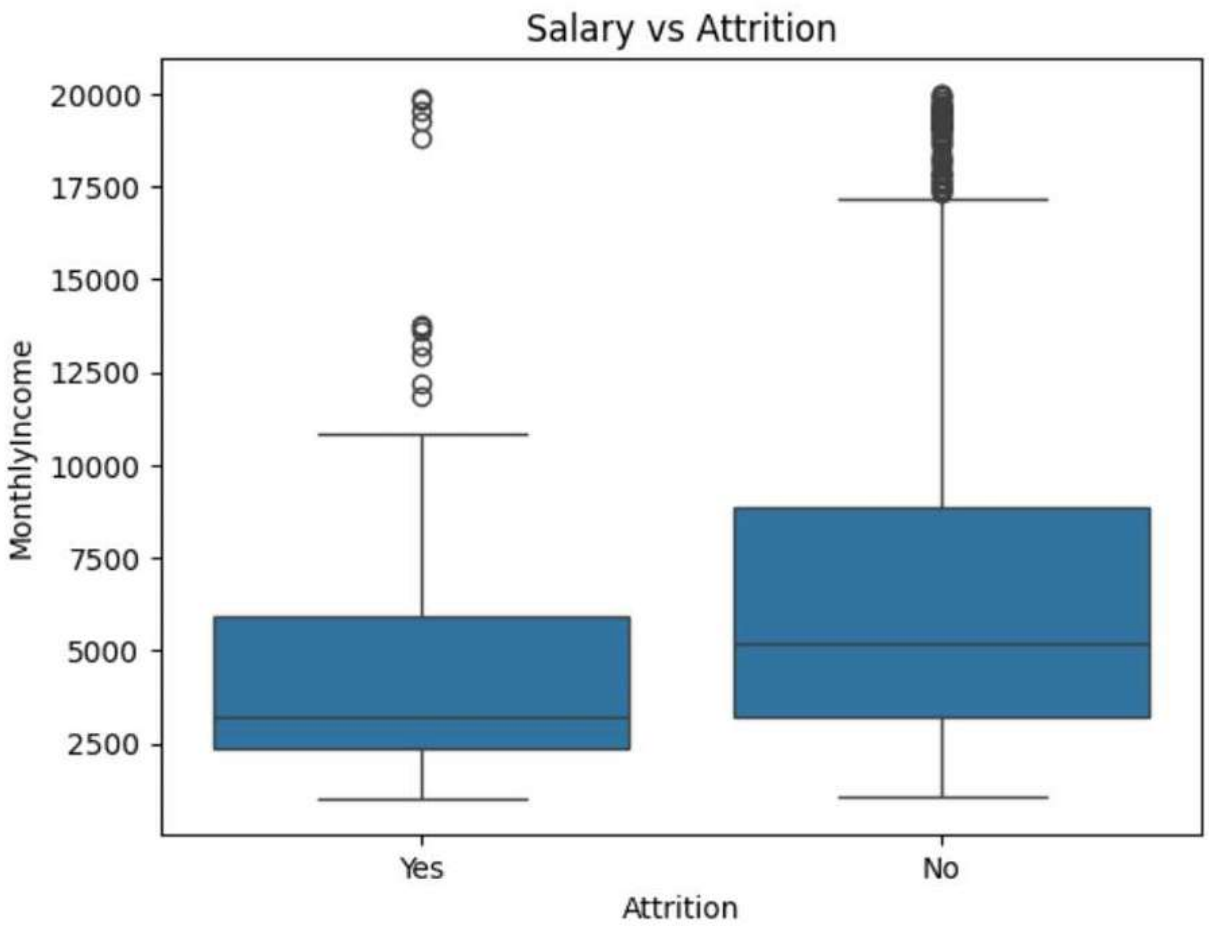


```python
plt.figure()
sns.boxplot(x='Attrition', y='MonthlyIncome', data=df)
plt.title("Salary vs Attrition")
plt.show()
```

[18] Python



Insight:
While leavers have a slightly lower median income, the distributions overlap heavily. Compensation alone does not explain attrition — non-financial factors matter more.

```python
pivot = pd.pivot_table(df, values='Attrition_Flag',
                       index='JobSatisfaction',
                       columns='OverTime')

plt.figure()
sns.heatmap(pivot, annot=True, cmap='Reds')
plt.title("High Risk: Overtime vs Low Satisfaction")
plt.show()
```

[17] Python

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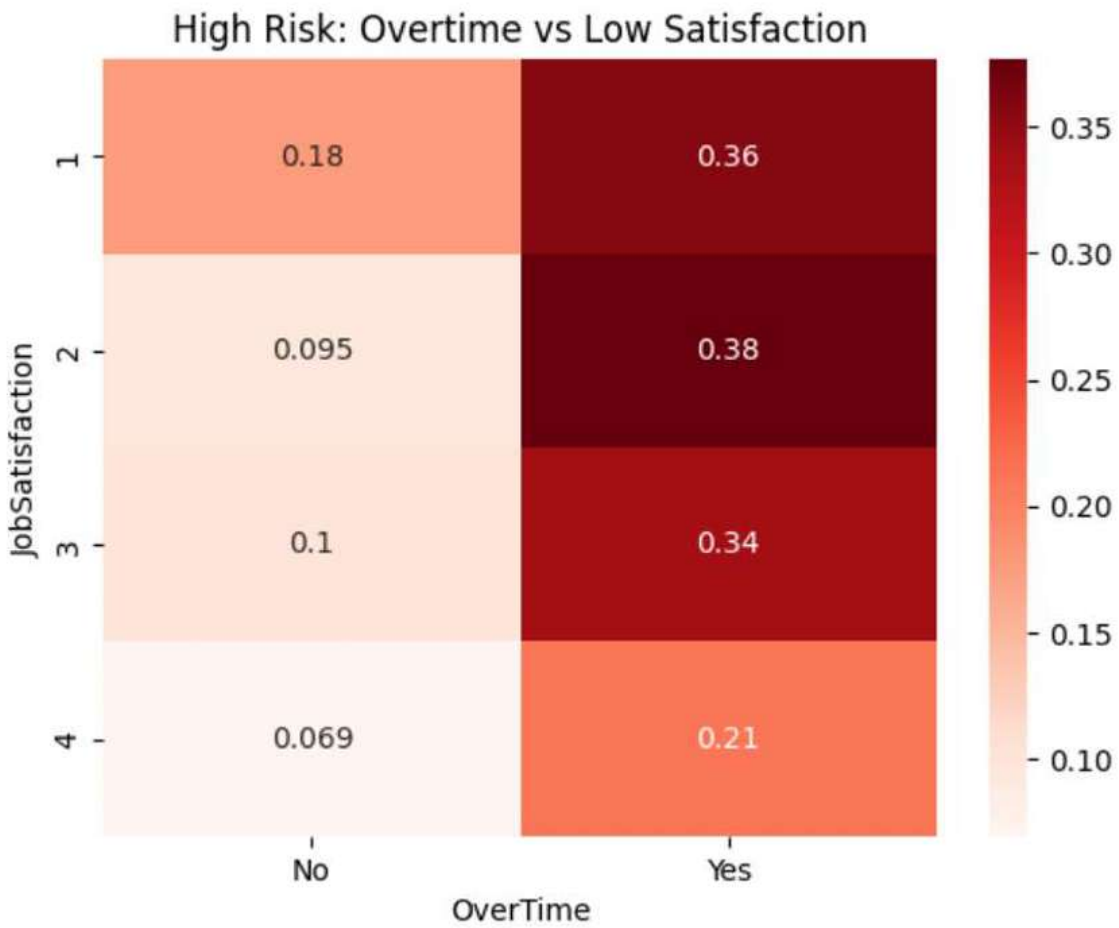
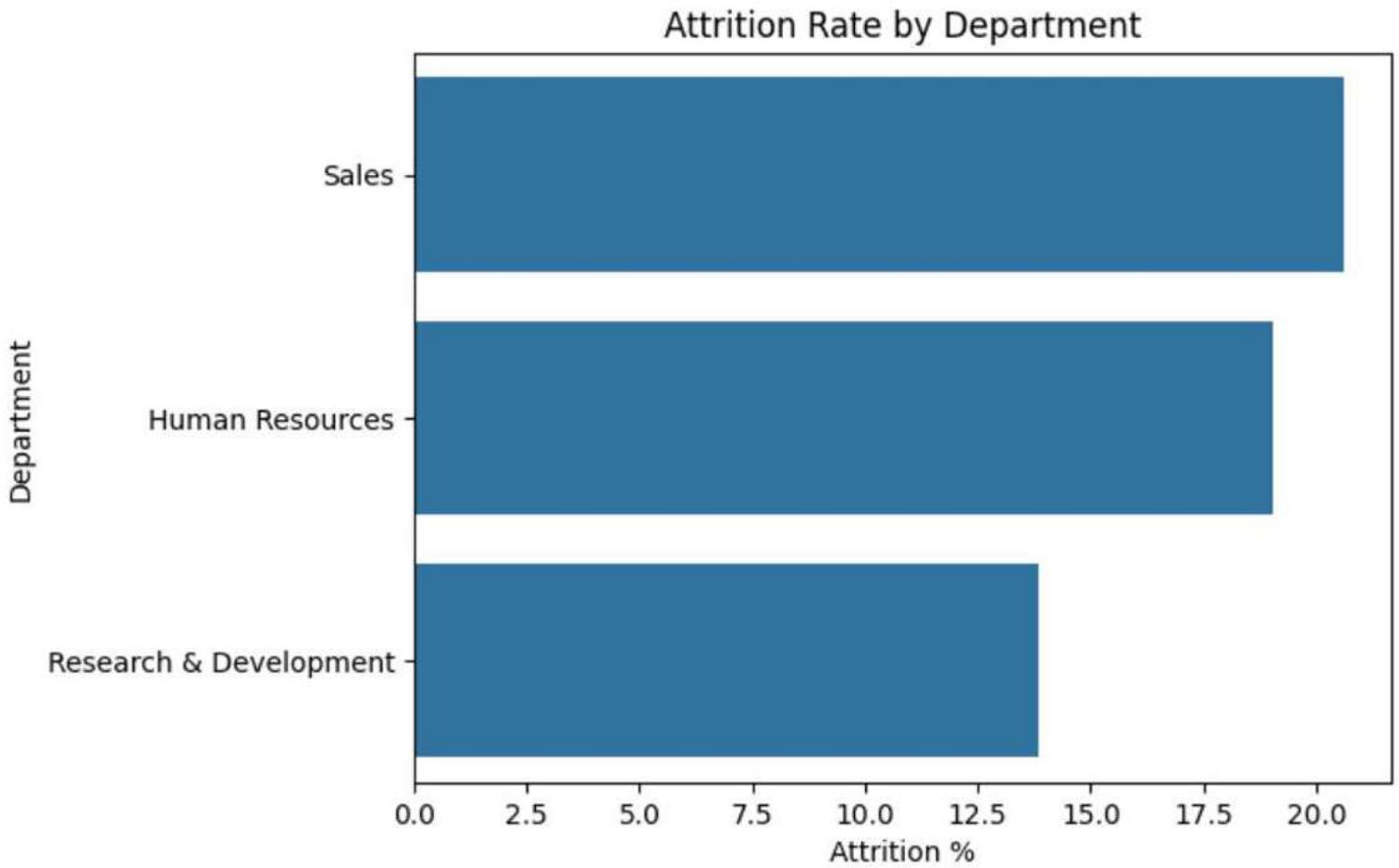


Insight:
Attrition jumps to 36% when an employee works overtime and has low job satisfaction. Without overtime and with high satisfaction, it drops to 7%. Overtime is a powerful multiplier of risk.

```python
dept = df.groupby('Department')['Attrition_Flag'].mean().sort_values(ascending=False)*100

plt.figure()
sns.barplot(x=dept.values, y=dept.index)
plt.title("Attrition Rate by Department")
plt.xlabel("Attrition %")
plt.show()
```

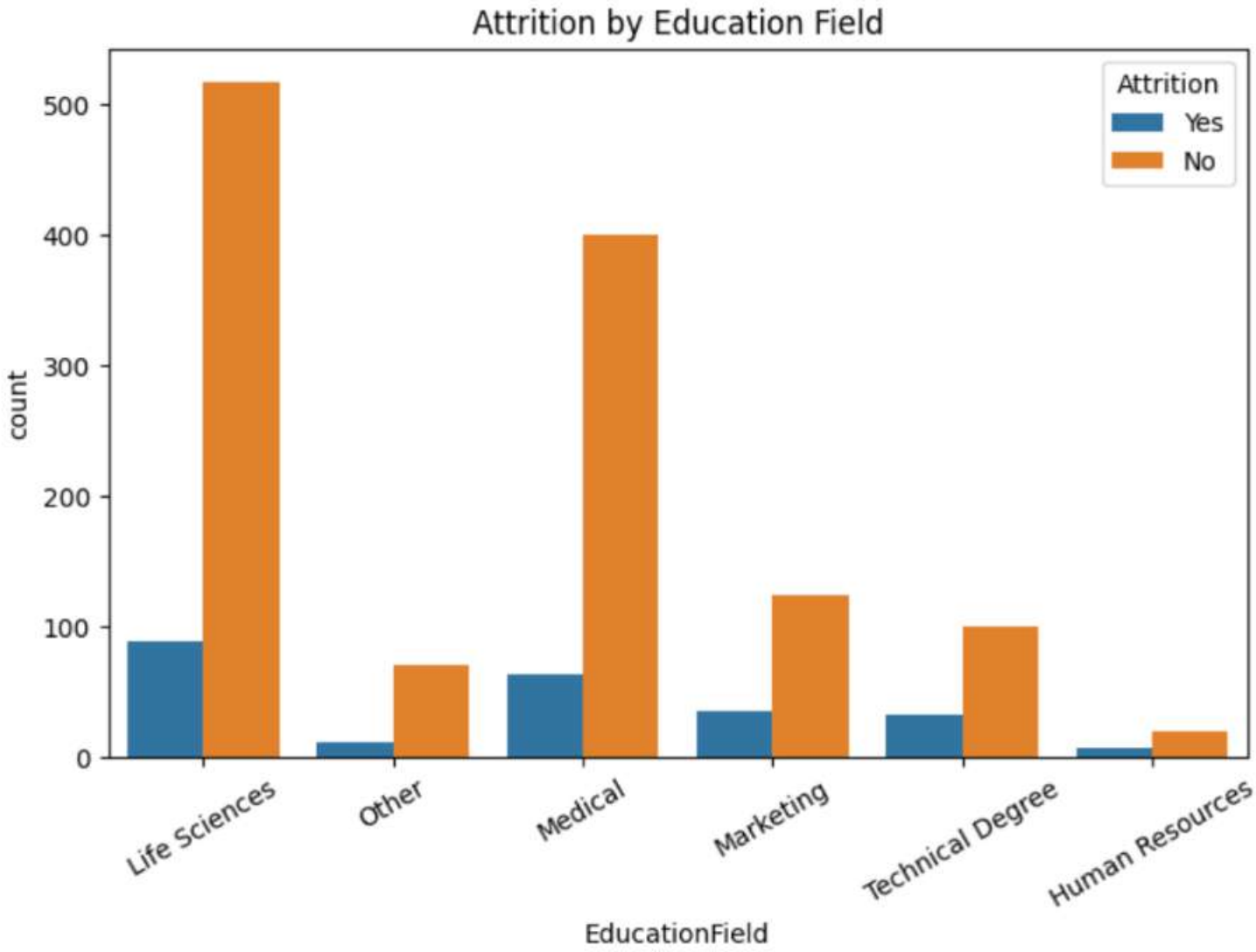
[16] Python



Insight:
Sales has the highest attrition (~20%), followed by HR (~18%). R&D is lowest (~8%). Department-specific strategies are needed, especially in Sales.

```python
plt.figure(figsize=(8,5))
sns.countplot(x='EducationField', hue='Attrition', data=df)
plt.xticks(rotation=30)
plt.title("Attrition by Education Field")
plt.show()
```

[4] ✓ 0.8s Python



Insight:
Attrition is highest among Life Sciences and Medical employees, while Marketing and Technical roles show moderate risk, indicating that attrition is concentrated in specific education domains rather than evenly distributed.